EVERSEEN COMPUTER VISION AI HACKATHON 2026 · PROBLEM STATEMENT 4

TaazaScan

See the rot before it shows.

A low-cost visible + near-infrared camera station that predicts each fruit's remaining shelf life and hidden bruises days before spoilage is visible, then tells Indian stores exactly what to do with it.

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EVERY FRUIT GETS AN ACTION



Sell first
First-expiring-first-out, not FIFO



Mark down today
Before it spoils, not after



Juice / cut-fruit counter
Rescue value from short-life fruit



Reject at receiving
Hidden bruises, with evidence for the supplier

Working proof of concept: 14× more about-to-spoil fruit caught before it looks bad; store waste cut from 9.7% to 1.8% in simulation

India loses ₹1.53 lakh crore of harvest every year

Spoilage starts inside
Cell-wall breakdown, fungal growth and bruising begin under the skin. When rot is visible, it is already unsellable.

FIFO and best-before dates are blind
Two crates bought on the same day can have very different lives. Dates and arrival order can't tell them apart.

Indian heat, thin cold chain
Produce sits at 30 °C+ in stores, mandis and dark stores. As a rule of thumb, ripening runs 2–3× faster per 10 °C.

Today's camera AI only sees the surface
Normal (RGB) cameras spot rot once it is visible — the same moment a staff member would.

₹1.53 lakh cr

value of India's post-harvest food losses, about 2.35% of GDP

NABCONS study, 2022

15% · 11.6%

of guava and tomato harvests lost after harvest — among the highest of any crop

NABCONS study, 2022

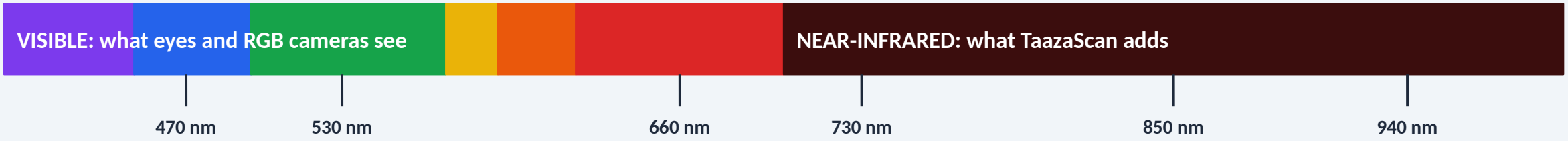
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of all food produced worldwide is wasted; fresh produce, meat and poultry lose the most

Everseen PS4 brief

The real question isn't "is it rotten?" It's "how many days does it have left?"

The insight: near-infrared light sees what eyes can’t



Our 6 LED wavelengths

What NIR reveals

- Water loss and cell-wall breakdown change 850 / 940 nm reflectance
- Chlorophyll loss shifts the 660 / 730 nm red edge
- Sub-surface bruises show up dark in NIR long before the skin changes

Proven in research

- Low-cost NIR imaging found 100% of slight apple bruises right after impact and within 3 hours (IJECE study)
- Hyperspectral imaging + deep learning predicts avocado and kiwi ripeness (Varga et al., 2021)

Our twist: multispectral on a budget

- Research hyperspectral cameras are very expensive
- We use a camera with its IR-cut filter removed + an LED ring flashing 6 wavelengths in sequence
- Result: a 6-band image in under a second, with commodity parts

How TaazaScan works: scan, predict, act



1. Receive
Crate arrives at the store, dark store or mandi.

2. Scan
Tray goes under the station: 6-band image + temperature log.

3. Predict
Days left for every fruit + hidden-bruise map, with confidence.

4. Decide
Compares days left with how fast that item sells.

5. Act
WhatsApp task in Kannada, Hindi or English; price label updated.

Sell first

First-expiring-first-out. Short-life fruit goes to the front of the display.

Mark down today

Only fruit that won't sell at full price before it spoils — no needless discounts.

Juice / cut-fruit

Fruit with under half a day left goes to the juice or cut-fruit counter.

Reject / claim

Hidden bruises at receiving: reject the lot or claim credit, with the NIR image as proof.

Food safety first: TaazaScan never puts spoiled food back on sale. It only acts earlier on fruit that is still good.

Architecture and models: from light to a store decision

SENSE

- Camera with IR-cut filter removed + LED ring: 470, 530, 660, 730, 850, 940 nm
- RGB camera
- Temperature & humidity logger; optional ethylene / VOC sensor
- Raspberry Pi 5 or Jetson at the edge

PERCEIVE

- White-reference calibration → true reflectance
- YOLOv8-seg: one mask per fruit
- Per-fruit spectral features: band ratios, red-edge index, NIR texture

PREDICT

- Hidden-bruise model (pixel map → fruit score)
- Remaining-shelf-life regressor with confidence band
- Temperature kinetics (Q10) to project life in a cold room vs the shop floor

DECIDE

- Sell-through aware: days left vs days until it sells
- Sell first / markdown / juice / reject rules
- ₹ value at risk per lot

ACT

- WhatsApp tasks in local languages
- Shelf price-label update
- Store dashboard
- Supplier quality scorecard

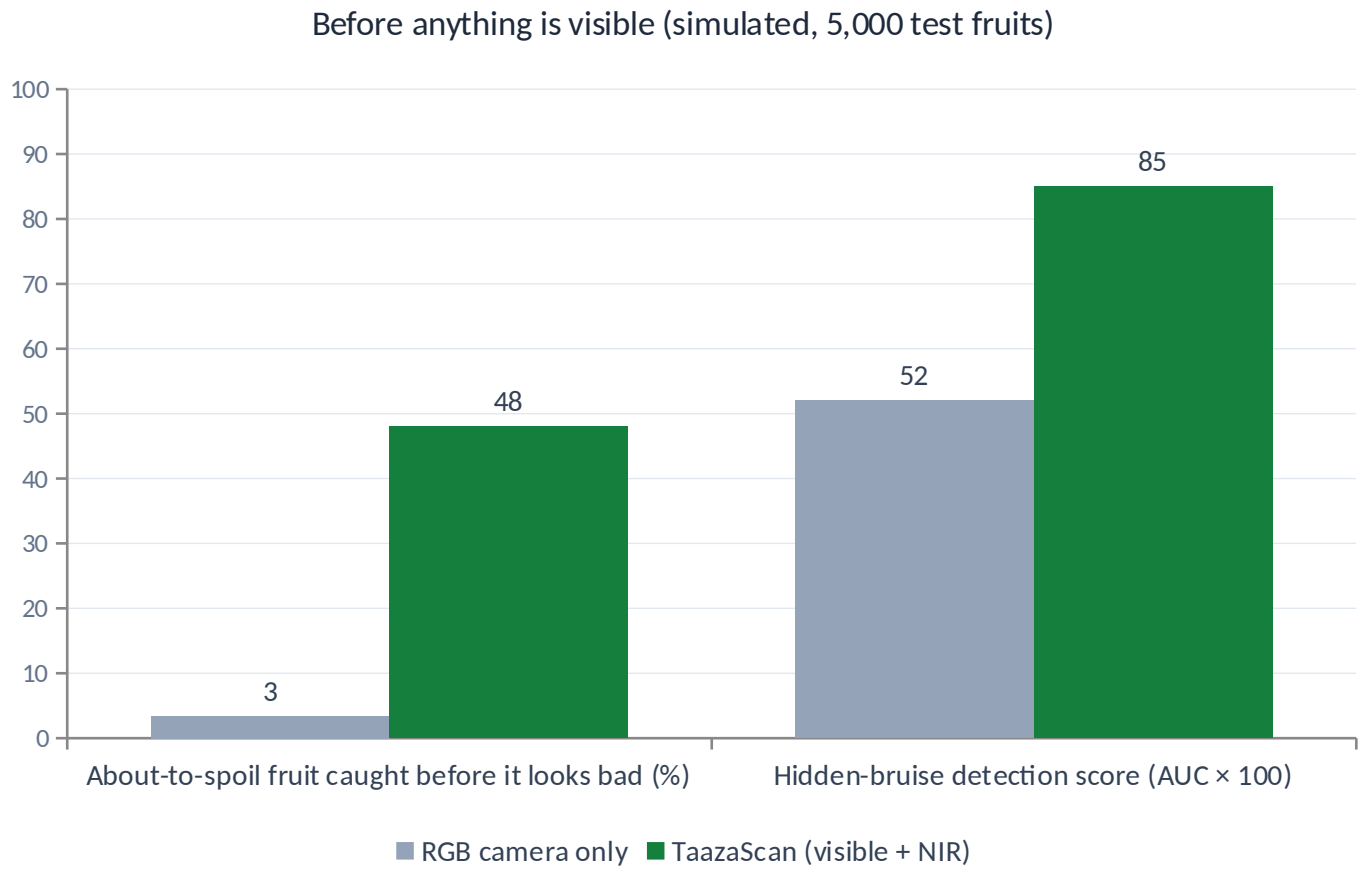
Built and tested now: prediction + decision core (highlighted). Finale: real station, real captures.

Low cost
Commodity camera, LEDs and a single-board computer. Target under ₹15,000 per station (estimate).

Works offline
Runs on the edge box, so patchy store internet doesn't stop it. Only results go to the cloud.

Easy to extend
Same pipeline for every fruit; a new fruit needs a short ripening capture to calibrate.

Proof of concept: TaazaScan sees what RGB misses



14× more about-to-spoil fruit flagged before it looks bad (48.0% vs 3.4% with RGB)

–81% store waste in a 30-day simulation at 30 °C (9.7% → 1.8% of fruit received)

0.41 d average error in predicted days left (RGB: 0.60 d). Wrong “sell now” flags: 0.2%

Simulation with a physics-based ripening model (banana, tomato, mango, guava; 25% hidden bruises; Q10 temperature effect). An RGB camera with the same decision rules gets waste to 3.0%; NIR cuts it further to 1.8%. Code and reproducible results: see the GitHub repository submitted with this deck.

Built for India: who uses it, what it saves, how it scales

Modern trade
Supermarket chains with fresh sections, e.g. DMart, Reliance Fresh / Smart, More, Spencer's

Quick-commerce dark stores
10-minute delivery hubs, e.g. Blinkit, Zepto, Swiggy Instamart, where returns for bad produce hurt

Mandis, FPOs and exporters
Objective grading at the gate: better prices for good lots, proof for rejections

Cold chain and warehouses
Decide which lots go to which city first, by days left

Brand names show target segments only; no partnership is implied.

ILLUSTRATIVE STORE ROI

≈ ₹79,000 / month

For a store selling ₹10 lakh of fresh produce a month:
waste 9.7% → 1.8% (our simulation) saves about ₹9.5 lakh a year, against a station costing under ₹15,000 (est.).

Pays back in weeks

Cheap hardware, big monthly saving

Less landfill methane

Less food waste, more food donated

Fair to suppliers

Image proof ends grading disputes

Finale plan: a working TaazaScan station in 48 hours

H 0–12

Build the station

Pi + no-IR-filter camera + 6-LED ring; white-reference calibration

→

H 12–24

Train on real data

Our ripening captures + FruitNet (Indian fruits) + DeepHS hyperspectral data

→

H 24–36

Decide and alert

Decision engine (done) + WhatsApp alerts + store dashboard

→

H 36–48

Live demo

Scan fruit live: days left, bruise map, action and ₹ saved

Starting now: a 7-day ripening time-lapse of bananas, tomatoes and guavas, scanned daily, so we arrive at the finale with real labelled data.

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AR — **Aravind Reddy Kunareddy** — App, alerts & dashboard

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